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COMMERCIAL ITEM DESCRIPTION

Diesel-Engine-Driven Pneumatic Tired Trenchers

This commercial item description covers self-propelled diesel-engine-driven pneumatic tired trenchers.

Type I - 45HP

Type II - 70HP

Salient characteristics:

A. Design. The trencher shall be pneumatic tire mounted, four wheel drive, four wheel power steered, air/liquid cooled, diesel engine driven, with heavy duty rear differential and designed to perform underground construction. The trencher shall be compatible with optional attachments or accessories which perform backfilling, spoils conveyance, trenching, rock sawing, backhoeing and cable laying. The operator's seat shall swivel 45 and 90 degrees, be adjustable forward and back and have a mechanical positive pin lock. The type I trencher shall 'be equipped with 31X15.50-15 8 ply bar lug tires. The type II trencher shall be equipped with 38x18-20 8 ply bar lug tires.

B. Engine. The engine shall be heavy duty air or liquid cooled, either two or four stroke cycle diesel. The engine shall be equipped with an efficient fuel injection mechanism, fuel filter system, air cleaner of the oil bath or dry type filter, and a lubricating oil filter. As a minimum, the engine shall have a Net Brake Horsepower, obtained and corrected in accordance with SAEJ1349, of not less than 45 hp for a type I trencher and not less than 70 hp for a type II trencher.

C. Ground drive. Ground speed while working or mobiling must be through

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hydrostatic drive with infinitely variable speed control using four selectable speed ranges in both forward and reverse. The ground drive motor shall have two speed ranges which can be shifted while mobiling.

D. Use conditions. The trenching machine and specified attachments shall negotiate 20 percent side slopes and 20 percent longitudinal slopes without the engine or other pressure lubricated components losing constant operating oil pressure and be able to operate without malfunction. These slope conditions shall be accomplished without any wheel of the trenching machine losing contact with the ground.

E. Temperature conditions. Without preheating from an external source, the engine shall start within five minutes and shall operate under full load conditions within 15 minutes after engine start. The engine shall be capable of operating in 15 to 115 degrees Fahrenheit ambient temperatures. The use of starting aids is permissible.

F. Safety. The trencher shall not start unless an operator is in the seat, the speed/directional control is in neutral, and the attachment drive control is in neutral. While operating, the trenching machine shall shut down if the operator leaves the seat and the speed/directional control or the attachment drive control is not in neutral. The four post type Roll-over Protection Structure (ROPS) shall be provided. The trencher(s) shall conform to all Federal Laws and Regulations governing noise, air pollution and other health and safety requirements which are in effect for this type of equipment at the time of contract award.

G. Hydraulic System. The trencher shall have an efficient, cool-running hydraulic system taking full advantage of a high capacity oil cooler, and a hydraulic oil reservoir with baffles.

H. Instrumentation and controls. The following instruments and controls, convenient to and in full view of the operator's normal operating position shall be furnished:

1. Voltmeter (or Ammeter)
2. Tachometer
3. Hourmeter
4. Fuel gauge
5. Oil pressure indicator
6. Air cleaner indicator
7. Speed/directional controls both foot and hand operated

I. Attachments. The trencher shall be capable of being equipped with any of the following attachments:

1. Multi-position digging attachment*
2. Vibratory plow with fiber optic package
3. Backhoe (24 inch bucket required, capable of digging

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at a minimum depth of 7 feet)
4. Hydraulic reel carrier
5. 72 inch 6 Way Backfill blade
6. Digging Chain, Type I - 50,000 pound Tensile Strength
Type II - 70,000 pound Tensile Strength

* A hydraulically movable attachment that allows trenching from centerline to full offset to the right-hand side and intermediate points of the trencher. The positioning of the digging unit can be accomplished from the operator's seat. (Other attachments shall be provided when specified.]

Boom to dig 52 inches in depth, when Type I is required, or 70 inches in depth when Type II is required, with capability of replacing the Boom tube only. Dual auger shall be 15 inches in length out each side as a minimum. The trencher shall be equipped with a mechanical trench cleaner and a 50,000 pound digging chain for Type I trencher or a 70,000 chain for a Type II.

Unless otherwise specified a multi-position digging attachment, a backhoe and a six way backfill blade shall be provided with each trencher.

J. Attachment drive. Modular attachments shall be driven by a hydrostatic pump with infinitely variable speed control. There shall be three styles of interchangeable gear boxes. The chain speeds for each gearbox shall be as follows:

Low 0 to 500 fpm
Standard 0 to 600 fpm
High 0 to 900 fpm

K. Identification plate. An identification plate, permanently marked in accordance with MIL-STD-130, shall be securely attached to the vehicle in a readily accessible location. The identification plate shall contain the following information:

NOMENCLATURE

MANUFACTURER'S MODEL NUMBER

MANUFACTURER'S SERIAL NUMBER

MANUFACTURER'S NAME

MANUFACTURER' S ADDRESS

MANUFACTURER' S SERVICE PHONE NUMBER

DATE OF MANUFACTURE

CONTRACT NUMBER

NATIONAL STOCK NUMBER

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REGISTRATION NUMBER

L. Trailer. Unless otherwise specified a trailer shall be provided for each trencher as required in this Commercial Item Description.

1. Design. The trailer shall be pneumatic tire mounted, tandem axle, flat bed unit with beaver tail and loading ramps. The trailer shall have an overall length of between 26 and 30 feet. The vehicle shall comply with all Federal Motor Carrier Safety Regulations applicable to the type of trailer furnished and in effect on the date of contract award.

a. Rated Capacity, The trailer shall have a fronted mounted jack rated at not less than 30,000 pounds and have a Gross Vehicle Weight Rating of not less than 26,000 pounds and a Gross Axle Weight Rating of not less than 20,000 pounds per axle.

b. Standard Vehicle and Accessories. The trailer, components, assemblies and accessories to be delivered under this contract shall be the manufacturer's standard or optional items, which ever is greater and meet the requirements of this specification.

2. Height.

a. Lunette. The trailer shall be fitted with a 3 inch non-swivel type lunette in accordance with MS 51336. The lunette height shall be adjustable from 22 to 32 in three inch or smaller increments. Height adjustments shall be possible by two mechanics using common hand tools in 15 minutes or less. Two 3/8 inch safety chains with a grab hook on the free end of each, shall be provided.

b. Trailer Bed. The bed height shall be between 33 and 36 inches when level. Design shall permit the trailer to load and off load with a minimum of two inches clearance maintained between the trencher equipped with any attachments (trailer bed level, on hard level surface). Two inches of clearance above the ground shall also be maintained during on/off loading.

3. Landing Gear. The trailer shall have a front mounted jack rated at not less than 30,000 pounds, The landing legs shall be equipped with self-leveling skid pads. Clips for holding the crank, when folded, shall be provided. The landing gear shall withstand, without deformation, the combined static and dynamic forces due to the proportion of gross weight sustained and forces resulting from impact during coupling and uncoupling operations. The landing gear assembly shall be mounted to plates that are gusseted to resist forward and rearward dynamic thrust on the landing leg. When placed in the travel position, the landing gear leg shall remain positively locked. The landing gear shall be protected to prevent the entrance of foreign matter which would impair its functional or mechanical efficiency. The maximum cranking effort to raise the fully loaded trailer shall not exceed 100 pounds.

4. Ramps. The trailer shall be equipped with a full width ramp or

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adjustable ramps capable of loading wheeled equipment with tire spaces as narrow as 30 inches to as wide as the width of the trailer. Ramp angle shall not exceed 11 degrees.

5. D-Rings. The trailer shall have a minimum of two D-Rings equally spaced down each side. Each lashing ring and its mounting shall be capable of withstanding pulls of at least 10,000 pounds in any direction without deformation.

6. Floor. 1 3/4 inches minimum dimensions pressure treated lumber raised not less than 1/4 inch above beam level.

7. Electrical. Lighting and wiring shall be in accordance with the applicable Federal Motor Carrier Safety Regulations. 12 volt I.C.C. standard equipment with a seven way connector receptacle conforming to SAE J560, with connectors connected and color coded as specified there in, shall be provided. Exterior lights shall be bulb replaceable resealable type with Lexan, turtle-back lens. Tail lights shall be Lexan, factory sealed. Combination turn/clearance light(s) shall be standard. Clearance and marker lights, reflectors and license plate light shall be standard.

8. Brakes.

a. Service" Brakes. The brakes shall be full air, internal expanding type and shall conform to the applicable Federal Motor Carrier Safety Regulations. The braking system shall include standard breakaway features, relay emergency valve, air reservoirs, piping, hose connections, and all other components required for a complete air brake system.

b. Parking brakes. The trailer shall be equipped with spring or air diaphragm mechanical lock type parking brakes. The parking brakes shall hold the trailer, with rated pay load, on a 10 percent incline up or down despite the depletion of the compressed air supply, with no additional application. The parking brake shall be automatically applied upon disconnection of the emergency air line and under emergency conditions. The parking brakes shall conform to the applicable Federal Motor Carrier Safety Regulations.

9. Tires. Tires shall be pneumatic, tubeless type, highway tread (7.50x16). Tires shall be steel belted radial type. The tires shall have a rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with the trailer loaded with the rated payload.

10. Trailer Data Plate. A trailer data plate, permanently marked in accordance with MIL-STD-130, shall be securely attached to the vehicle in a readily accessible location. The data plate shall contain, as a minimum, the same information as the identification plate listed above.

11. Finish. All steel components shall be treated to retard rust and painted using the manufacturer's standard paint and color.

M. The vehicle furnished must be the latest model as of the date of award

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(with minor modifications, as that term is defined in DoD FAR Supplement 252.211-7012, Paragraph K-85 of the solicitation) of an established product previously produced and sold in substantial commercial quantities {exceeding the criteria at FAR 15.804-39(f)(2)(ii)(A)) for at least three years immediately preceding the issuance date of the solicitation. The vehicle furnished, or earlier models of it, absent minor modifications, shall have routinely been supported with spare/repair parts which were produced or sold in the normal course of business. At the time of delivery, the contractor shall furnish the Administrative Contracting Officer with verification of compliance with these requirements.

NOTE TO OFFERORS: Offerors shall provide, as part of their proposals, the following information on the vehicle (or its earlier models) being offered as a commercial item:

(1) Total sales to the U.S. Government or to contractors for U.S. Government use during the three years immediately preceding the issuance date of the solicitation.

(2) Total sales of the item to the general public during the three years immediately preceding the issuance date of the solicitation.

(3) Length of time the item has been sold in the commercial market place.

(4) Points of contact and phone numbers of customers to which the item has been sold in the commercial market place.

(5) Other pertinent information necessary to determine that the item is an established commercial item.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Recovered materials. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and

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exterior package quantities? labeling, and marking shall be as specified in the contract and/or order. Identification marking and data plates shall be in accordance with MIL-STD-1223.

NOTES

a. The type trencher required shall be specified.

b. Unless otherwise specified trencher(s) shall be painted contractor's standard commercial color.

c. Commercial manuals to include parts manual, and operation and maintenance manuals shall be supplied with each trencher. (Two copies of each shall be supplied).

d. Unless otherwise specified a trailer shall be provided with each Trencher as required in this Commercial Item Description.

e. Additional attachments shall be as specified.

Custodian:
AIR FORCE-99

Preparing Activity:
AIR FORCE-84

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